



IPG - SPECIFICATION DOCUMENT

Bank 2 wallet / B2C

Version 1.9

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Billing & IT – Mobile Money and VAS

Version History

Version	Date	Changed By	Change summary
1.0.	17-Jul-18	Christian Belewete	Initial draft
1.1	20-Jul-18	Christian Belewete	Adding Service flow
1.2	24-Jul-18	Christian Belewete	Adding Login request parameters
1.3	10-Aug-18	Christian Belewete	Adding test details
1.4	17-Jun-19	Christian Belewete	Updating error codes
1.5	18-Jun-19	Christian Belewete	Updating error codes
1.6	28-Oct-19	Christian Belewete	Updating error codes
1.7	08-June-20	Christian Belewete	Updating urls and headers
1.8	29-Jun-21	Christian Belewete	Updating urls
1.9	14-Feb-22	Christian Belewete	Updating iPG AA URLs



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1 Introduction

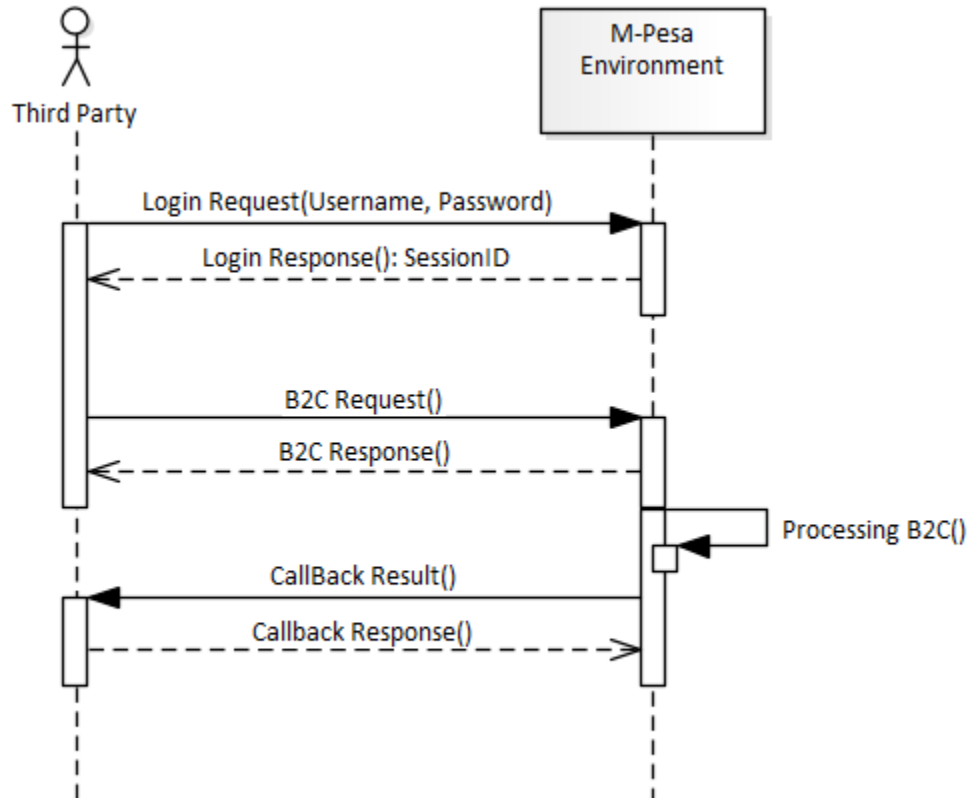
The iNSight Payment Gateway sits in front of G2 which manages a customer's wallet. This wallet, in the case of mobile network operators, the "Mobile Money Platform" is physically separated from the payment gateway (sometimes also referred to as a "Broker" or an "Orchestration Layer") providing the benefit of a simplified interface and abstraction of both function and security.

1.1 Prerequisites

Below items are required before for System Owner of iNSight Payment Gateway to proceed with Third-Party onboarding

#	ITEMS	OWNER
1	Ensure end to end connectivity	VDRC / NETWORK TEAM
2	Third-Party Identification details : - Name - Contacts - ShortCode - CommandID	VDRC / M-COMMERCE TEAM
3	Third-Party to configure Callback URL for notification (This service must run on Port 443 or 3000)	THIRD-PARTY
4	Third-Party user creation	VDRC / PAYMENT GATEWAY OWNER

2 Service flow



1. The third party initiates the process by sending the Login Request
2. On a successful login, M-Pesa respond with a **SessionID**
3. Third Party system sends **B2C Request**, using **SessionID** received in step 2 as **token**. M-Pesa will send a acknowledge response.
4. After processing the request, M-Pesa will forward the final result of to the Third Party Callback Url sent on the B2C Request.
5. The Third Party has to send a response with the required format.



3 Connection details

3.1 Test

Login	
URL Login	https://uatipg.m-pesa.vodacom.cd:8091/insight/SOAPIn
URL B2C	https://uatipg.m-pesa.vodacom.cd:8094/iPG/B2C
WAF IP	192.168.115.57
Public IP	41.78.195.169
Event ID Login	2500
Event ID B2C	12001

3.2 Production

Login	
URL Login	https://ipg-aa.m-pesa.vodacom.cd:8091/insight/SOAPIn
URL B2C	https://ipg-aa.m-pesa.vodacom.cd:8094/iPG/B2C
WAF IP	192.168.115.245
Public IP	41.78.195.183
Event ID Login	2500
Event ID B2C	12001

4 API Calls Parameters

4.1 Login

4.1.1 Request Parameters

Name	Notes	Type
Header		
EventID	<i>ID of the event to be actioned on Transaction Switch. This is a fixed value for Login of 2500</i>	Integer Required
Request Data Items		
Username	<i>Username to validate</i>	String Required
Password	<i>Password of the user to validate</i>	String Required

4.1.2 Response Parameters

Name	Notes	Type	Sample Values
Event Info			
code	<i>iNSight process status code</i>	Integer	3, 4
description	<i>Description of the iNSight process status code</i>	String	Processed, Error
detail	<i>Additional information about iNSight process</i>	String	Processed, Authentication Failed
Request Data Items			
Username	<i>Username that was evaluated</i>	String	Same as value sent in request
Password	<i>Password of the user that was evaluated</i>	String	Same as value sent in request
Response Data Items			
SessionID	<i>In the case of a valid login, the session id to be used as the Token ID for future event calls</i>	String	da6be0443e53411ebd90ad0aec7a97ab

4.2 B2C

4.2.1 Request Parameters

Name	Notes	Type
Header		
Token	<i>Token received from a Login request</i>	String Required
EventID	<i>ID of the event to be actioned on Transaction Switch.</i>	Integer Required
Request Data Items		
ServiceProviderName	<i>Name of the service provider. For example: FINCA</i>	String Required
CustomerMSISDN	<i>MSISDN of the customer</i>	String Required
Currency	<i>Currency Code for the transaction. For example: CDF or USD</i>	String Required
Amount	<i>Amount of the said currency to transfer from the Third-Party to the customer</i>	String Required
TransactionDateTime	<i>Date of the transaction. Format YYYYMMDD HH24MISS.</i>	Date Optional
Shortcode	<i>Shortcode of the Third-Party in M-Pesa where funds will move from.</i>	String Required
CustomerAccountNumber	<i>Customer's Third-Party account number.</i>	String Optional
Language	<i>Customer's language choice.</i>	String Required
ThirdPartyReference	<i>The Third-Party's Transaction ID.</i>	String Required
CallBackChannel	<i>The channel used to send the results of the transaction to the third party. 1. Synchronous (default and only available if Mobile Money system is synchronus) 2. Async HTTP Post 3. Async SMS 4.Async eMail</i>	String Required
CallBackDestination	<i>Dependant on CallBackChannel Should be populated with relevant URL, MSISDN (format example 243821111111) or email address</i>	String Required
CommandID	<i>M-Pesa CommandID. M-Pesa team to provide.</i>	String Required
InitiatorIdentifier	<i>Third-Party's Identifier. M-Pesa team to provide.</i>	String Optional
SecurityCredential	<i>Third-Party's Password. M-Pesa team to provide.</i>	String Optional

4.2.2 Response Parameters

Name	Notes	Type	Sample Values
Event Info			
Code	iNSight process status code	Integer	3, 4
Description	Description of the iNSight process status code	String	Processed, Error
Detail	Additional information about iNSight process	String	Authentication Failed, Session Expired
TransactionID	Transaction ID from the payment gateway	String	
Request Data Items			
ServiceProviderName	String		Same as value sent in request
CustomerMSISDN	String		Same as value sent in request
Currency	String		Same as value sent in request
Amount	String		Same as value sent in request
TransactionDateTime	Date		Same as value sent in request
Shortcode	String		Same as value sent in request
CustomerAccountNumber	String		Same as value sent in request
Language	String		Same as value sent in request
ThirdPartyReference	String		Same as value sent in request
ServiceProviderName	String		Same as value sent in request
CallBackChannel	String		Same as value sent in request
CallBackDestination	String		Same as value sent in request
CommandID	String		Same as value sent in request
InitiatorIdentifier	String		Same as value sent in request
SecurityCredential	String		Same as value sent in request
Response Data Items			
Insight_txid	The unique transaction ID from iPG	String	12sdxq3634dadsas242
ResponseCode	Code indicating the result of the transaction. 0 – Received request successfully. -1 – Transaction will not be processed. Try again later.	String	0, -1, -2

5 API Calls Samples

5.1 Login

5.1.1 Request

```
POST /insight/SOAPIn HTTP/1.1
URL: https://uatipg.m-pesa.vodacom.cd:8091/insight/SOAPIn
Host: uatipg.m-pesa.vodacom.cd
Content-type:text/xml; charset=utf-8

<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope
  xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:soap="http://www.4cgroup.co.za/soapauth"
  xmlns:gen="http://www.4cgroup.co.za/genericsoap">
  <soapenv:Header>
    <soap:EventID>2500</soap:EventID>
  </soapenv:Header>
  <soapenv:Body>
    <gen:getGenericResult>
      <Request>
        <dataitem>
          <name>Username</name>
          <type>String</type>
          <value>{{ USERNAME }}</value>
        </dataitem>
        <dataitem>
          <name>Password</name>
          <type>String</type>
          <value>{{ PASSWORD }}</value>
        </dataitem>
      </Request>
    </gen:getGenericResult>
  </soapenv:Body>
</soapenv:Envelope>
```

5.1.2 Response

```
<?xml version='1.0' encoding='UTF-8'?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/"
  <S:Header>
    <ns3:eventid
      xmlns:ns2="http://www.4cgroup.co.za/genericsoap"
      xmlns:ns3="http://www.4cgroup.co.za/soapauth">2500
    </ns3:eventid>
  </S:Header>
  <S:Body>
    <ns2:getGenericResultResponse
      xmlns:ns2="http://www.4cgroup.co.za/genericsoap"
      xmlns:ns3="http://www.4cgroup.co.za/soapauth">
      <SOAPAPIResult>
        <eventInfo>
          <code>3</code>
          <description>Processed</description>
          <detail>Processed</detail>
          <transactionID>0c49ec4bbebe44b082aa24e5f548b514</transactionID>
        </eventInfo>
        <request>
          <dataItem>
            <name>Username</name>
            <type>String</type>
            <value>thirdpartyc2bw</value>
          </dataItem>
          <dataItem>
            <name>Password</name>
            <type>String</type>
            <value>thirdpartyc2bw</value>
          </dataItem>
        </request>
        <response>
          <dataItem>
            <name>SessionID</name>
            <type>String</type>
            <value>dbcb63a40afd4191a990d66ed9cb62f7</value>
          </dataItem>
        </response>
      </SOAPAPIResult>
    </ns2:getGenericResultResponse>
  </S:Body>
</S:Envelope>
```

5.2 B2C

5.2.1 Request

```
POST /iPG/B2C HTTP/1.1
URL: https://uatipg.m-pesa.vodacom.cd:8094/iPG/B2C
Host: uatipg.m-pesa.vodacom.cd
Content-type:text/xml; charset=utf-8

<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope
  xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header>
    <soap:EventID xmlns:soap="http://www.4cgroup.co.za/soapauth">12001</soap:EventID>
    <soap:Token xmlns:soap="http://www.4cgroup.co.za/soapauth">{{ TOKEN }}</soap:Token>
  </soapenv:Header>
  <soapenv:Body>
    <gen:getGenericResult
      xmlns:gen="http://www.4cgroup.co.za/genericsoap">
      <Request>
        <dataltem>
          <name>ServiceProviderName</name>
          <type>String</type>
          <value>{{ ServiceProviderName }}</value>
        </dataltem>
        <dataltem>
          <name>CustomerMSISDN</name>
          <type>String</type>
          <value>{{ MSISDN }}</value>
        </dataltem>
        <dataltem>
          <name>Currency</name>
          <type>String</type>
          <value>USD</value>
        </dataltem>
        <dataltem>
          <name>Amount</name>
          <type>String</type>
          <value>1</value>
        </dataltem>
        <dataltem>
          <name>TransactionDateTime</name>
          <type>Date</type>
          <value>20180601123700</value>
        </dataltem>
        <dataltem>
          <name>Shortcode</name>
          <type>String</type>
          <value>15058</value>
        </dataltem>
      </Request>
    </gen:getGenericResult>
  </soapenv:Body>
</soapenv:Envelope>
```



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```
</dataItem>
<dataItem>
  <name>Language</name>
  <type>String</type>
  <value>EN</value>
</dataItem>
<dataItem>
  <name>ThirdPartyReference</name>
  <type>String</type>
  <value>Christian-Test1</value>
</dataItem>
<dataItem>
  <name>CallBackChannel</name>
  <type>String</type>
  <value>2</value>
</dataItem>
<dataItem>
  <name>CallBackDestination</name>
  <type>String</type>
  <value>https://callbackurl</value>
</dataItem>
<dataItem>
  <name>CommandID</name>
  <type>String</type>
  <value>InitTrans_one4allb2c</value>
</dataItem>
</Request>
</gen:getGenericResult>
</soapenv:Body>
</soapenv:Envelope>
```

5.2.2 Response

```
<?xml version='1.0' encoding='UTF-8'?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <S:Header>
    <ns3:eventid
      xmlns:ns2="http://www.4cgroup.co.za/genericsoap"
      xmlns:ns3="http://www.4cgroup.co.za/soapauth">12001
    </ns3:eventid>
  </S:Header>
  <S:Body>
    <ns2:getGenericResultResponse
      xmlns:ns2="http://www.4cgroup.co.za/genericsoap"
      xmlns:ns3="http://www.4cgroup.co.za/soapauth">
      <SOAPAPIResult>
        <eventInfo>
          <code>3</code>
          <description>Processed</description>
          <detail>Processed</detail>
          <transactionID>74b2f36481524d1e9bf907edbc86291d</transactionID>
        </eventInfo>
        <request>
          <dataItem>
            <name>ServiceProviderName</name>
            <type>String</type>
            <value>ONE4ALL</value>
          </dataItem>
          <dataItem>
            <name>CustomerMSISDN</name>
            <type>String</type>
            <value>243811835361</value>
          </dataItem>
          <dataItem>
            <name>Currency</name>
            <type>String</type>
            <value>USD</value>
          </dataItem>
          <dataItem>
            <name>Amount</name>
            <type>String</type>
            <value>1</value>
          </dataItem>
          <dataItem>
            <name>TransactionDateTime</name>
            <type>Date</type>
            <value>20180601123700</value>
          </dataItem>
        </request>
      </SOAPAPIResult>
    </ns2:getGenericResultResponse>
  </S:Body>
</S:Envelope>
```



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```
<dataItem>
  <name>Shortcode</name>
  <type>String</type>
  <value>15058</value>
</dataItem>
<dataItem>
  <name>Language</name>
  <type>String</type>
  <value>EN</value>
</dataItem>
<dataItem>
  <name>ThirdPartyReference</name>
  <type>String</type>
  <value>Christian-Test1</value>
</dataItem>
<dataItem>
  <name>CallBackChannel</name>
  <type>String</type>
  <value>2</value>
</dataItem>
<dataItem>
  <name>CallBackDestination</name>
  <type>String</type>
  <value>https://callbackurl</value>
</dataItem>
<dataItem>
  <name>CommandID</name>
  <type>String</type>
  <value>InitTrans_one4allb2c</value>
</dataItem>
</request>
<response>
  <dataItem>
    <name>Insight_txid</name>
    <type>String</type>
    <value>lurzsmmaxquh9l1svold6e3hx7x7jp9t</value>
  </dataItem>
  <dataItem>
    <name>ResponseCode</name>
    <type>String</type>
    <value>0</value>
  </dataItem>
</response>
</SOAPAPIResult>
</ns2:getGenericResultResponse>
</S:Body>
</S:Envelope>
```

5.2.3 Callback Result

To receive the result of an Asynchronous event call via an HTTP post, the CallBackChannel must be set to “2”. The CallBackDestination should also carry the URL that should be posted to from iNSight (it is important that iNSight can access this URL). The Third-Party must reply with the specified Response packet.

Request sent to the third party

```
POST /insight/SOAPIn HTTP/1.1
URL: http://xxx.xxx.xxx.xxx:aaa/insight/SOAPIn
Host: xxx.xxx.xxx.xxx
Content-type:text/xml; charset=utf-8

<?xml version="1.0" encoding="utf-8"?>
<soapenv:Envelope
  xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:soap="http://www.4cgroup.co.za/soapauth"
  xmlns:gen="http://www.4cgroup.co.za/genericsoap">
  <soapenv:Header>
    <soap:EventID>1</soap:EventID>
  </soapenv:Header>
  <soapenv:Body>
    <gen:getGenericResult>
      <Request>
        <dataItem>
          <name>ResultType</name>
          <value>0</value>
          <type>String</type>
        </dataItem>
        <dataItem>
          <name>ResultCode</name>
          <value>0</value>
          <type>String</type>
        </dataItem>
        <dataItem>
          <name>ResultDesc</name>
          <value> Process service request successfully.</value>
          <type>String</type>
        </dataItem>
        <dataItem>
          <name>OriginatorConversationId</name>
          <value>fbb550d9eb924e8e99ac1aa7ff8c05d0</value>
          <type>String</type>
        </dataItem>
        <dataItem>
          <name>ConversationId</name>
```

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```
<value>0c9766a5-1c9e-4ffa-b815-3b447dde5cb1</value>
<type>String</type>
</dataItem>
<dataItem>
  <name>TransactionID</name>
  <value>X-BS281</value>
  <type>String</type>
</dataItem>
<dataItem>
  <name>Amount</name>
  <value>200</value>
  <type>String</type>
</dataItem>
<dataItem>
  <name>TransactionTime</name>
  <value>20170301T22:59:01</value>
  <type>String</type>
</dataItem>
<dataItem>
  <name>ThirdPartyReference</name>
  <value>123456-Test</value>
  <type>String</type>
</dataItem>
<dataItem>
  <name>InsightReference</name>
  <value>12sdxq3634dadsas242</value>
  <type>String</type>
</dataItem>
</Request>
</gen:getGenericResult>
</soapenv:Body>
</soapenv:Envelope>
```

Response

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<response>
  <dataItem>
    <name>ResponseCode</name>
    <type>String</type>
    <value>200</value>
  </dataItem>
  <dataItem>
    <name>RESULT_CODE</name>
    <type>String</type>
    <value>Received</value>
  </dataItem>
</response>
```

6 Error Reference

6.1 Error

6.1.1 ResponseCode (Synchronous response)

CODE	DESCRIPTION	COMMENT
0	Processed	Your request has been received
-2	Duplicate Transaction	ThirdPartyReference should be unique for each request

6.1.2 ResultCode (Callback result)

CODE	DESCRIPTION	COMMENT
0	Process service request successfully	Payment successful
-1	System internal error.	
1000	System is busy, the service request is rejected.	
1001	Caller authentication error.	Caller incorrect credentials used
1002	Parameter missing or data type error.	
1003	Duplicate originator conversation ID.	
2000	Service code error.	
2001	Initiator authentication error.	Incorrect credentials used
2002	Receiver invalid.	
2003	Currency invalid.	Currency allowed are : USD/CDF
2004	Amount invalid.	Integer for CDF. Float with 2 decimals for USD
2005	Rule limited.	
2006	Balance insufficient.	
2007	Conversation ID invalid.	
2010	Request has been rejected by the approver.	
2011	Request process expired because of approval timeout.	
2023	File load error.	
2036	One and only one of the query parameter ReceiptNumber, OriginatorConversationID and ConversationID can appear at the same time.	
2051	MSISDN invalid.	
2052	KYC field invalid.	
2053	Trust level invalid.	

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2054	Account Type invalid.
2055	New operator ID invalid.
2056	New PIN invalid.
2057	Language code invalid.
2058	Voucher invalid
2059	Product invalid.
2060	Product mismatched.
2061	ShortCode invalid or already used by another identity.
2062	TillNumber invalid or already used by another identity.
2063	OperatorID invalid or already used by another identity.
2064	UserName invalid or already used by another identity.
2065	OperatorRole invalid or not in available range.
2066	The mandatory KYC is missing.
2100	MSISDN should be absent when Receiver is a Till.
2101	MSISDN is inconsistent with Receiver MSISDN.
2102	RegisterBy should be absent when initiated by OrgOperator.
2103	Products of Till cannot be all deleted.
2104	Receiver should be absent when initiated by OrgOperator.
2105	O2CLink number invalid.
2106	Duplicate O2CLink number.
2107	Duplicate TaxCode.
2108	The Caller was Suspended by CustomerCare.
2109	StartTime cannot be later then EndTime.
2110	Duplicate verification question.
2111	Transaction status invalid.
2112	Only transactions can be queried by SearchTransaction.
2113	OriginatorConversationID or ConversationID invalid, the request does not exist.
2114	The request being queried cannot be a query operation.
2115	Initiator cannot perform an operation on the self.
2116	The identity is not suspended by Customer Care.
2117	The Original state cannot be changed to new state.
2118	The PIN is not locked.
6100	The corresponding point account does not exist.
6101	The receiver is not supported for the operation.



6102	The loyalty point account balance is insufficient.
6103	The identity account to be created already exists.
6104	Multiple loyalty point accounts are matched based on then input parameter.
6105	This adjustment category is incorrect.
6106	The amount format is incorrect.
6108	The amount can not be 0.
6109	Field length exceeds the maximum.
6110	Can not find the loyalty point account.
6111	No loyalty point account log is found.
6112	Incorrect input parameters.
6113	The corresponding PointType is not found in the system.
6114	The corresponding PointType is not in Released state.
6115	The loyalty point account type found in the system is not applicable to the current identity.
6116	The date format is incorrect.
6117	Loyalty points fail to be queried.
6118	The amount can not be 0.
6119	The identity does not have any loyalty point account, Failed to query the loyalty points.
2801	Invalid Image Size
2802	Invalid File Type